Constraining Black Hole Growth in the JWST Era

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Abstract

Aims. We assess which JWST-identified accreting black holes and candidates at $z \geq 4$ place meaningful constraints on early growth histories, and how these constraints depend on seed formation, radiative efficiency, and reported mass uncertainty.

Methods. We compile a provenance-aware catalogue containing 350 literature measurements across 32 source families, represented by 338 object records in 337 host systems pending final identity reconciliation. We compare the 237 objects with eligible numerical black-hole masses against analytic growth models spanning seed and accretion assumptions; the other 101 remain explicit no-inference records. We propagate published asymmetric mass errors using 10,000 deterministic Monte Carlo draws and preserve mass-method systematics as separate envelopes. Catalogue products and 676 per-object panels reproduce through an ordered five-notebook workflow.

Results. Under the reference assumptions $z_{\rm seed} = 30$, $\epsilon = 0.1$, no merger boost, and a $10^2\,M_\odot$ seed, 14 objects require a lifetime-average $f_{\rm Edd} > 1$ at their point estimates. Ten remain above unity at the 16th percentile, and eight have $P(f_{\rm Edd} > 1) \geq 0.95$. UNCOVER-20466 has the highest required average rate, with median $f_{\rm Edd} = 1.453$ and a 16th–84th percentile interval of 1.354–1.549. At the ideal non-spinning thin-disk efficiency $\epsilon = 0.05719$, no point estimate requires $f_{\rm Edd} > 1$ under otherwise unchanged assumptions. The inferred growth requirements are therefore conditional on the adopted model; heterogeneous selection functions and mass estimators preclude interpreting global rankings as demographic measurements.

Key words. black hole physics – accretion, accretion disks – galaxies: active – galaxies: high-redshift – quasars: supermassive black holes

1 Introduction

The discovery of massive black holes in the early Universe raises a fundamental question: what growth histories could have assembled them within the available cosmic time? The strength of this constraint depends on both the observed black-hole mass and redshift, as well as assumptions about seed formation, radiative efficiency, and the duration of active accretion. An apparently demanding object under one set of assumptions may therefore be compatible with less extreme growth under another.

Here, we use a literature-based catalogue of JWST-identified high-redshift accreting black holes and candidates to ask which objects meaningfully constrain early growth histories. The atlas covers $z \geq 4$, extending from the first billion years to a cosmic age of approximately 1.54 Gyr at its lower-redshift boundary. We calculate the average accretion rates required across a range of seed and growth assumptions, incorporating reported mass uncertainties and distinguishing measurements with different evidential and methodological limitations. Source-native measurements, identity links, and mass provenance are retained to support comparison across heterogeneous literature samples.

Our aim is to identify which growth requirements remain demanding across plausible scenarios, which depend primarily on uncertain assumptions, and where improved observations would most strengthen the constraints. This approach treats the catalogue as a means of testing possible growth histories, rather than assuming that every high-redshift black hole presents the same assembly problem. The constraints apply to individual objects under stated assumptions; we do not estimate a pooled mass function, occurrence rate, or formation-channel fraction.

The input literature spans JWST broad-line samples [3, 4, 5, 6, 7, 8, 11, 12, 13, 14, 15, 16, 17, 18, 35, 36], host-selected broad-line candidates [10], an X-ray evidence history [19, 21], JADES narrow/high-ionization candidates [22], and GN-z11 high-ionization evidence [23]. Heterogeneous additions include high-ionization and narrow-line candidates [24, 25, 26, 30, 31], MIRI SED-selected candidates [27, 28], GHZ9 [29], compact blue broad-line emitters [32], additional UNCOVER candidates [33], MoM-BH*-1 [34], and GHZ4/GHZ7 [37].

1.1 Theoretical framework

The growth constraints follow from the relation between accretion luminosity, retained mass, and available cosmic time. For electron-scattering opacity in ionized hydrogen, the Eddington luminosity and luminosity-based Eddington ratio are

$$L_{\rm Edd} = \frac{4\pi G M_{\rm BH} m_p c}{\sigma_T}, \qquad f_{\rm Edd} = \frac{L_{\rm bol}}{L_{\rm Edd}}, \qquad t_{\rm Edd} = \frac{M_{\rm BH}c^2}{L_{\rm Edd}} = \frac{c\sigma_T}{4\pi G m_p} \simeq 0.45 \text{ Gyr}. \tag{1}$$

Here G, c, m_p, and σ_T denote the gravitational constant, speed of light, proton mass, and Thomson cross section. With radiative efficiency $0 < \epsilon < 1$ and rest-mass inflow rate $\dot{M}_{\rm in}$,

$$L_{\rm bol} = \epsilon \dot{M}_{\rm in} c^2, \qquad \dot{M}_{\rm BH} = (1-\epsilon)\dot{M}_{\rm in} = \frac{1-\epsilon}{\epsilon}\frac{f_{\rm Edd}}{t_{\rm Edd}} M_{\rm BH}. \tag{2}$$

The inflow rate is defined at the radiating flow; additional mass loss before that point is not modelled. At fixed efficiency, define

$$\Delta t = t(z_{\rm obs}) - t(z_{\rm seed}) > 0, \qquad \overline{f_{\rm Edd}} = \frac{1}{\Delta t}\int_{t(z_{\rm seed})}^{t(z_{\rm obs})} f_{\rm Edd}(t)\, dt. \tag{3}$$

The analytic growth relation adopted by Dayal [2] then becomes

$$M_{\rm BH}(t_{\rm obs}) = B_{\rm merge} M_{\rm seed} \exp\left[\frac{1-\epsilon}{\epsilon}\frac{\overline{f_{\rm Edd}}\,\Delta t}{t_{\rm Edd}}\right], \qquad t_{\rm efold} = \frac{\epsilon}{1-\epsilon}\frac{t_{\rm Edd}}{f_{\rm Edd}}. \tag{4}$$

The e-folding time assumes constant positive $f_{\rm Edd}$; it is 0.05 Gyr at $\epsilon = 0.1$ and $f_{\rm Edd} = 1$. The dimensionless $B_{\rm merge}$ represents an effective multiplicative merger boost, not a resolved merger history.

Cosmic ages use the flat matter-plus-Lambda relation

$$t(z) = \frac{2}{3H_0\sqrt{\Omega_\Lambda}} \operatorname{asinh}\left[\sqrt{\frac{\Omega_\Lambda}{\Omega_m}}(1+z)^{-3/2}\right], \tag{5}$$

with $H_0 = 67.3\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_m = 0.315$, and $\Omega_\Lambda = 0.685$. The Hubble constant is converted to inverse time; ages and growth intervals are in Gyr. Radiation is neglected. In particular, the supplementary $z_{\rm seed} = 3400$ comparison extrapolates this age relation and is not a radiation-era seed-formation calculation.

Two inversions provide the principal growth diagnostics:

$$\overline{f_{\rm Edd}}_{\rm req} = \max\left[0, \frac{\epsilon}{1-\epsilon}\frac{t_{\rm Edd}}{\Delta t}\ln\left(\frac{M_{\rm BH}}{B_{\rm merge}M_{\rm seed}}\right)\right], \tag{6}$$

$$\log_{10}\left(\frac{M_{\rm seed,req}}{M_\odot}\right) = \log_{10}\left(\frac{M_{\rm BH}}{M_\odot}\right) - \log_{10} B_{\rm merge} - \frac{1-\epsilon}{\epsilon}\frac{\overline{f_{\rm Edd}}\,\Delta t}{t_{\rm Edd}\ln 10}. \tag{7}$$

In Eq. 7, $M_{\text{seed,req}}$ denotes the required seed mass. The zero floor in Eq. 6 follows the implementation: if the boosted seed already exceeds the observed mass, no positive accretion is required, but this does not make that fixed history an exact mass match.

For a two-state history with the same efficiency in both states,

$$\overline{f_{\text{Edd}}} = D f_{\text{burst}} + (1-D) f_{\text{quiet}}, \qquad D_{\text{req}} = \frac{\overline{f_{\text{Edd}}}_{\text{req}} - f_{\text{quiet}}}{f_{\text{burst}} - f_{\text{quiet}}}. \tag{8}$$

Here D is the active-state time fraction, $f_{\text{burst}} > f_{\text{quiet}} \geq 0$, and the diagnostic requires $\overline{f_{\text{Edd}}}_{\text{req}} \geq f_{\text{quiet}}$. Values $D_{\text{req}} > 1$ identify an infeasible specified two-state history and are retained. An average growth requirement is not an instantaneous Eddington ratio or a uniquely measured duty cycle.

For the illustrative spin-dependent maps, the ideal Kerr thin-disk efficiency is the ISCO binding efficiency [1]:

$$\epsilon_a = 1 - \sqrt{1 - \frac{2}{3 r_{\text{ISCO}}}}, \tag{9}$$

$$r_{\text{ISCO}} = 3 + Z_2 - \operatorname{sgn}(a)\sqrt{(3 - Z_1)(3 + Z_1 + 2Z_2)}, \tag{10}$$

$$Z_1 = 1 + (1 - a^2)^{1/3}\left[(1+a)^{1/3} + (1-a)^{1/3}\right], \qquad Z_2 = \sqrt{3a^2 + Z_1^2}. \tag{11}$$

The dimensionless spin is $a = cJ/(GM_{\text{BH}}^2)$, where J is the signed angular momentum relative to the disk, and r_{ISCO} is in units of GM_{BH}/c^2. The ideal efficiencies at $a = -1, 0, +1$ are approximately 0.038, 0.057, and 0.423, respectively, neglecting photon capture and stress at the ISCO. Above unity, the maps adopt a phenomenological luminosity ansatz,

$$f_{\text{Edd}} = 1 + \ln \dot{m}, \qquad \dot{m} = \frac{\dot{M}_{\text{in}}}{L_{\text{Edd}}/(\epsilon_a c^2)}, \qquad \epsilon_{\text{eff}} = \begin{cases} \epsilon_a, & f_{\text{Edd}} \leq 1, \\ \epsilon_a f_{\text{Edd}} \exp(1 - f_{\text{Edd}}), & f_{\text{Edd}} > 1. \end{cases} \tag{12}$$

The logarithmic luminosity ansatz applies only on the super-Eddington branch. This is an illustrative parameter-scan prescription, not a relativistic slim-disk or spin-evolution calculation. If efficiency varies with time, the integrated growth depends on $\int[(1-\epsilon(t))/\epsilon(t)] f_{\text{Edd}}(t)\, dt$; it cannot in general be represented by separately averaged efficiency and rate.

2 Catalogue construction

2.1 Scope and versioning

The catalogue uses scientific dataset versions rather than software releases. v1 is the original 23-object JADES broad-line catalogue. v2 expands the same mass-comparable analysis to 211 JWST broad-line objects. v3 retains all v2 objects and adds heterogeneous JWST-identified evidence classes. A source enters only one admission version; an internally heterogeneous source is assigned in full to v3. The fixed scope requires $z \geq 4$ and identification or candidate classification from JWST data. Objects merely followed up by JWST after a non-JWST discovery are outside the admission rule.

Source-family and measurement membership is frozen at the review cutoff of 3 September 2026; corrections to admitted measurements apply across affected versions, while new measurements require a declared subsequent version.

The v3 catalogue has 350 measurement rows, 338 physical objects, and 337 host systems. It contains 247 broad-line AGN, 49 narrow-line candidates, 30 photometric candidates, ten high-ionization candidates, and two X-ray candidates. Preferred object-level evidence comprises 214 secure, 41 probable, 82 candidate, and one disputed classification. These labels describe the source papers' evidence and do not place all classes on a single purity scale.

2.2 Provenance, extraction, and identity

Each measurement carries stable measurement, physical-object, and host-system identifiers. Source-native values, adopted standardized values, uncertainty semantics, table locations, URLs, DOI or archive versions, archive SHA-256 hashes where available, extraction dates, and caveat tags are retained. The manual-extraction audit covers 26 hash-pinned extraction artifacts, while the provenance registry contains 38 records covering all 32 admitted source families. Independent checks cover all 244 numerical masses and both error fields. Auxiliary coordinate and contextual sources are attributed separately to the THRILS catalogue [9], the UHZ1 spectroscopy paper [20], and the versioned JADES DR3 GOODS-S prism catalogue.

Identity resolution preserves every admitted measurement while selecting one preferred row for object-level inference. Alternate measurements are never silently averaged. The final completion illustrates this policy: 15 NEXUS NIRCam/WFSS measurements [35], 13 COSMOS-3D NIRCam-grism measurements [36], and GHZ4/GHZ7 [37] add 30 measurements but 29 physical objects. NEXUS source NX10835 lies 0.066 arcsec from an existing Mascia identifier at consistent redshift, so the new mass-bearing measurement becomes preferred for the same object.

The Scholtz et al. source audit provides another cardinality check. The paper reports 42 objects, whereas its source-native table contains 41 rows. Exactly 21 table rows satisfy $z \geq 4$; all are admitted. One is identity-linked to a broad-line object, yielding 20 distinct narrow-line candidates from that source. JADES-NS-GS00099671 is retained with its published host and bolometric observables but without an invented black-hole mass.

2.3 Mass comparability and analysis subsets

A measurement enters numerical growth analysis only when the source publishes a canonical numeric black-hole mass compatible with the declared comparison policy. Conditional masses inferred by assuming an Eddington ratio, indirect model ranges, and upper limits remain contextual observables. This produces 244 eligible measurements and 237 eligible preferred objects. Of those objects, 236 use Balmer-line single-epoch virial masses and one uses a UV single-epoch virial mass. Their redshifts span 4.000–10.603 with median 5.110; standardized $\log_{10}(M_{\rm BH}/M_{\odot})$ spans 5.51–9.31 with median 7.18. The 101 objects without an eligible mass remain visible throughout the catalogue and atlas. The stricter primary subset has 234 measurements and 227 preferred objects: it excludes candidate or disputed evidence, conditional mass interpretations, and methods not flagged as primary-comparable. Numerical eligibility alone therefore does not imply membership in the primary comparison subset.

Table 1: v3 scale and analysis subsets.

Product or subset	Rows or objects
Measurements / physical objects / host systems	350 / 338 / 337
Source-local observables	810
Growth-eligible measurements / objects	244 / 237
Primary-analysis measurements / objects	234 / 227
Catalogue-only physical objects	101
Alternate-measurement sensitivity pairs	7
Source-family caveat rows	32
Per-object parameter-map panels	676

3 Growth-model comparison

3.1 Reference model

We apply the relations in Section 1.1 to each eligible object. The reference ranking uses $z_{\mathrm{seed}} = 30$, $\epsilon = 0.1$, $B_{\mathrm{merge}} = 1$, and $M_{\mathrm{seed}} = 10^2\, M_\odot$. Required average rates and seed masses follow Eqs. 6 and 7. We call the derived requirement *growth pressure*: it measures how demanding a specified history is, conditional on the published mass and model. A single merger factor and fixed efficiency do not resolve feedback cycles, changing accretion states, or stochastic mergers.

The spin-dependent f_{Edd}–seed-mass maps and compatibility matrices use Eq. 12. The baseline rankings, duty-cycle diagnostics, and seed-redshift–mass maps retain $\epsilon = 0.1$; growth tracks retain their stated constant efficiencies. Thus compatibility above unity is conditional on a different efficiency model from the baseline required-f_{Edd} ranking.

3.2 Uncertainty and ranking policy

Published asymmetric mass uncertainties in $\log_{10} M_{\mathrm{BH}}$ are propagated with 10,000 deterministic split-side draws per measurement and per preferred object. If a source gives no reported mass error, the atlas does not invent one. In particular, the twelve mass-bearing NEXUS objects have no published mass errors. Their repeated point values yield zero-width stored intervals; probabilities and uncertainty ranks are left unavailable, not interpreted as certainty. They are marked separately in uncertainty figures. The other 225 objects have reported mass errors. Split-side sampling assigns half the probability to each side of the reported central value, using the respective error as the scale of a half-normal distribution; it approximates quoted intervals rather than reconstructing source posteriors. Redshifts are held fixed. Draws use seed 20260808 with a stable identifier-based random stream. For 10,000 draws the binomial Monte Carlo standard error of an estimated probability is at most 0.005 (approximately 0.0022 at probability 0.95); this numerical sampling error is separate from astrophysical uncertainty. Source-reported errors are retained as quoted and are not assumed to be purely statistical. For ZS7, the published 0.4-dex error already includes virial-calibration scatter. No additional scatter is added to that error. Separately reported method-scale systematics are stored and, where possible, mapped to sensitivity envelopes; the atlas does not assume that these terms are independent Gaussian errors.

The final published Killi mass $8^{+0.5}_{-0.4} \times 10^8 M_\odot$ is converted by transforming both linear bounds, giving $\log_{10}(M/M_\odot) = 8.90309^{+0.02633}_{-0.02228}$.

The primary scientific comparison is within object class and mass-comparability group. Global point and uncertainty ranks are retained only for navigation. The point navigation score is

$$S = \min\left[100, 100 \max\left(C\left(\frac{f_2 - 0.3}{1.2}\right), C\left(\frac{s_{0.3} - 4}{2.8}\right)\right) + 8C\left(\frac{z-6}{4}\right)\right], \tag{13}$$

where $C(x) = \min(1, \max(0, x))$, f_2 is the required f_{Edd} for a $10^2\, M_\odot$ seed, and $s_{0.3}$ is the required $\log_{10}(M_{\mathrm{seed}}/M_\odot)$ at $f_{\mathrm{Edd}} = 0.3$. This heuristic combines two growth diagnostics and a redshift term; it is not a probability. Ties use decreasing f_2, decreasing redshift, then stable identifier. The manuscript's required-f_{Edd} table is instead ordered directly by f_2. Compatibility is also calculated object by object across four seed families, three spin/efficiency cases, two merger cases, and three average accretion rates. For fixed $z_{\mathrm{seed}} = 30$, ϵ, B_{merge} and f_{Edd}, a cell is compatible precisely when the required $\log_{10}(M_{\mathrm{seed}}/M_\odot)$ lies inside the inclusive seed interval: $[1, 2]$ (light), $[3, 4]$ (intermediate), $[4, 6]$ (heavy), or $[2, 6]$ (PBH-labelled). These are overlapping scenario ranges, not probability priors. A required seed below the interval indicates that this fixed history would overgrow the observed mass; it does not exclude the seed family under a shorter duty cycle. The PBH label specifies only a mass range at the common starting redshift, not primordial formation physics or a PBH abundance constraint. The source-level selection registry leaves inverse-probability weights unset because no common inclusion-probability model spans the 32 source families. Consequently, compatibility fractions are descriptive summaries of the admitted objects, not population frequencies.

4 Results

4.1 Growth pressure under the reference assumptions

The required $f_{\rm Edd}$ distribution for a $10^2\,M_\odot$ seed has median 0.574 and interquartile range 0.488–0.680. Fourteen of 237 eligible objects require $f_{\rm Edd} > 1$ at their point estimates; none requires $f_{\rm Edd} > 2$. The high-pressure tail is not simply a mass ranking because less cosmic time can offset a lower observed mass. GN-z11, for example, has a lower standardized mass than many broad-line objects but has the second-largest required $f_{\rm Edd}$ at $z = 10.603$. Its UV virial estimator and high-ionization evidence place it outside the primary comparison subset; this position is a conditional cross-method diagnostic, not a claim of equivalent mass-estimator reliability.

UNCOVER-20466 leads both the point and uncertainty navigation views. Its point-estimate required $f_{\rm Edd}$ is 1.452, and its Monte Carlo median is 1.453 with a 16th–84th percentile interval of 1.354–1.549. The newly admitted COSMOS3D-13852 ranks third by the composite navigation score and fourth by required $f_{\rm Edd}$. GS-20057765 has the third-highest required $f_{\rm Edd}$. Table 2 reports the five largest point-estimate required $f_{\rm Edd}$ values, independent of the composite score.

Table 2: Highest point-estimate required $f_{\rm Edd}$ for a $10^2\,M_\odot$ seed under the reference assumptions. Intervals propagate published mass errors only. This navigation table spans mass methods; GN-z11 is excluded from the primary subset.

Object	z	$\log_{10}(M_{\rm BH}/M_\odot)$	Required $f_{\rm Edd}$	16th–84th percentile
UNCOVER-20466	8.500	8.17	1.452	1.354–1.549
GN-z11	10.603	6.20	1.437	1.335–1.542
GS-20057765	8.913	7.33	1.355	1.177–1.513
COSMOS3D-13852	7.646	8.60	1.314	1.247–1.384
RUBIES-EGS-55604	6.986	9.31	1.267	1.250–1.284

4.2 Robustness to reported mass uncertainty

The Monte Carlo results preserve the point-estimate conclusion for the strongest objects. Fourteen objects have median required $f_{\rm Edd} > 1$, ten remain above unity at the 16th percentile, and 15 cross unity by the 84th percentile. Eight objects have $P(f_{\rm Edd} > 1) \geq 0.95$. These probabilities are conditional on the source-reported errors and the reference growth assumptions; they do not add a common virial-calibration uncertainty or uncertainty in seed redshift, radiative efficiency, and merger history. Restricting to the 227 primary objects gives 12 point estimates above unity, eight above unity at the 16th percentile, and six with $P(f_{\rm Edd} > 1) \geq 0.95$. These are descriptive counts within the stated subset, not population fractions.

A separate publication-version check replaces the frozen Baccus v1 measurements for 44 exact-ID, position-confirmed counterparts with the published Table 1 values. Five frozen objects lack a published-table counterpart. Both retaining and omitting those five objects preserve the headline uncertainty counts above and the top five by required $f_{\rm Edd}$. The 32 changed central masses shift by -0.10 to $+0.04$ dex. This is a measurement-version sensitivity test with fixed evidence/method taxonomy; it does not constitute a

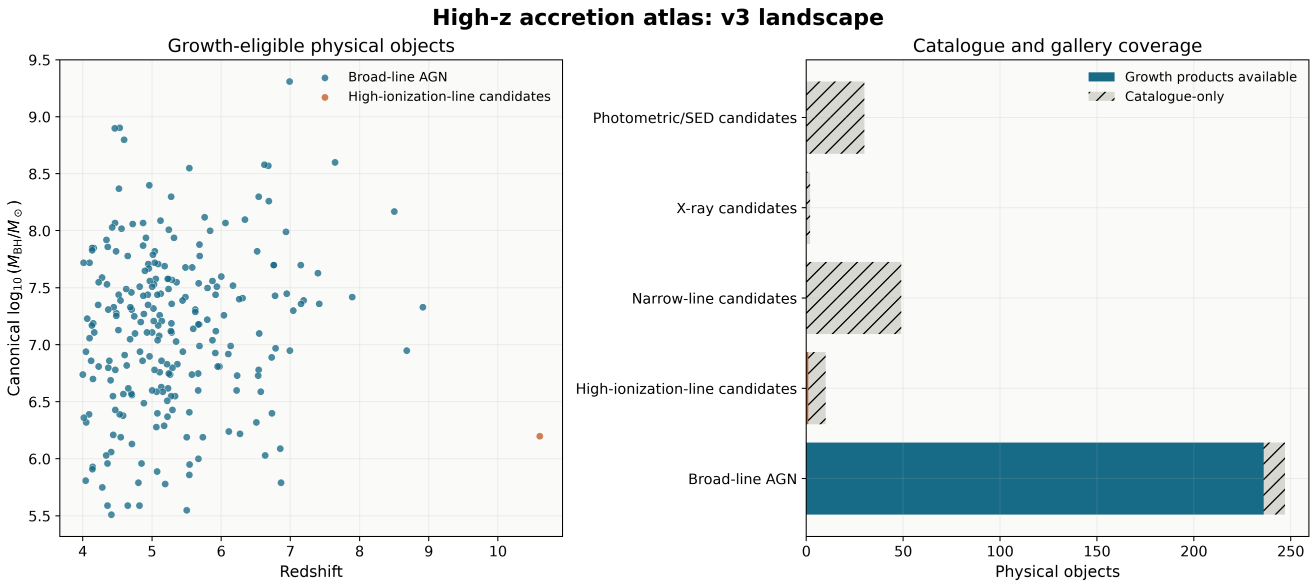

Figure 1: Catalogue landscape. Left: the 237 growth-eligible physical objects in black-hole-mass–redshift space. Right: product coverage for all 338 objects. The 101 catalogue-only objects remain available for provenance and evidence analysis without unsupported numerical growth products.

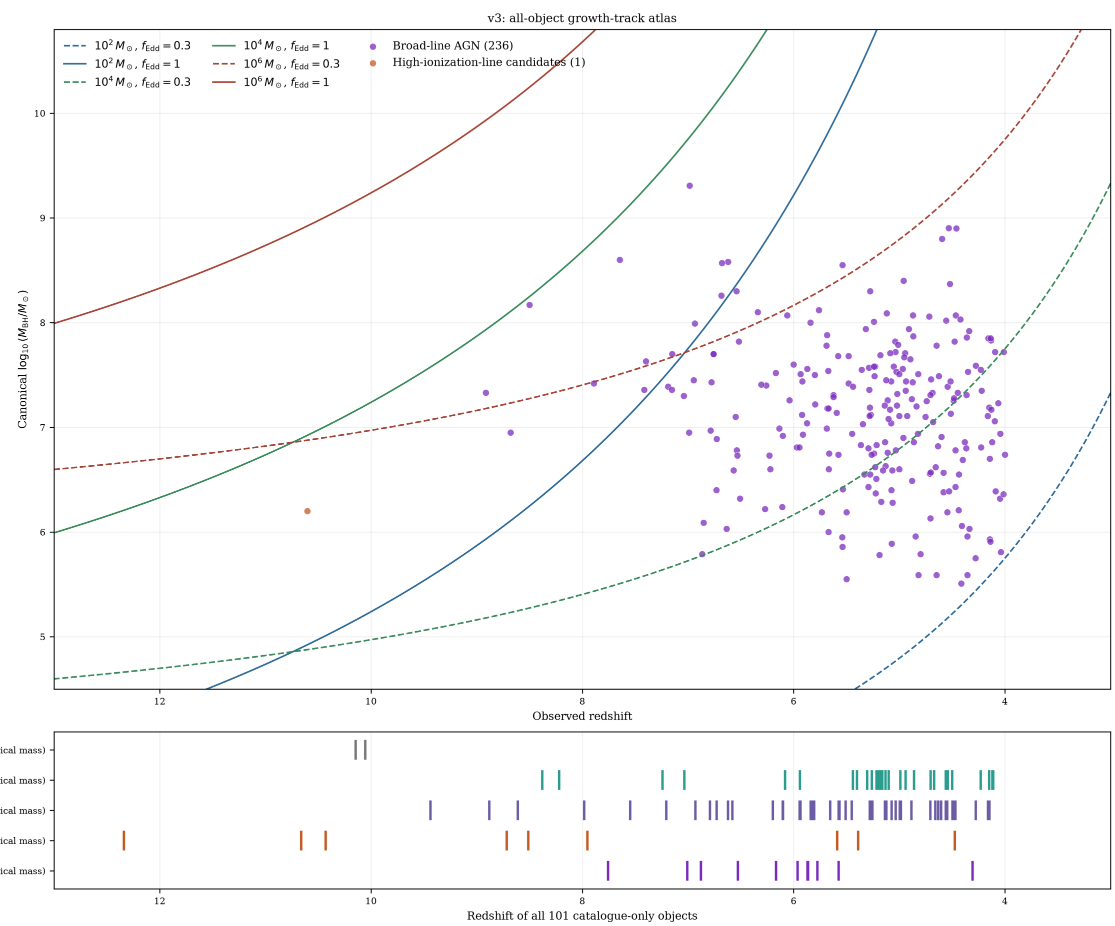

Figure 2: All-object growth-track view. The upper panel compares the 237 supported masses with reference tracks; the lower panel retains the redshift and class of all 101 no-mass objects. The reversed redshift axis spans $z = 13$ to $z = 3$.

new admission of the complete published sample. The object-level comparison and three-scenario summary are distributed with the v3 science tables.

4.3 Model compatibility and measurement choice

Compatibility changes substantially across the seed, efficiency, merger, and average-accretion grid. This dependence is the scientific point of the compatibility layer: an object's apparent tension cannot be assigned to seed mass alone without specifying the other assumptions. The complete matrix keeps unavailable cases distinct from incompatibility, preventing the 101 objects without eligible masses from being counted as failed growth histories.

Seven eligible objects have retained alternate measurements. Their sensitivity table reports how preferred-measurement choice changes standardized mass and required f_{Edd}. The follow-up matrix includes all 338 objects: ten are classified as robust high-pressure targets, seven as caveated high-pressure targets, 31 as uncertainty-leverage targets, 21 as comparison anchors, one as a source-consistency target, 167 as contextual records, and 101 as requiring a method-comparable mass before growth inference.

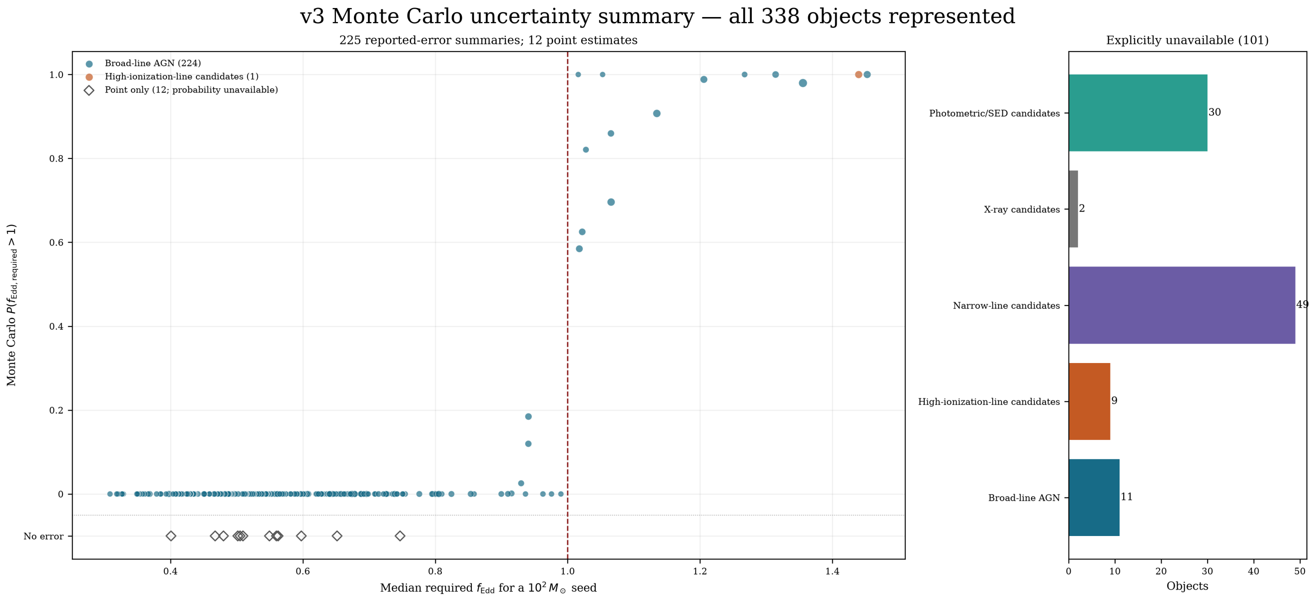

Figure 3: Uncertainty summary for 225 objects with reported mass errors and 12 point estimates without mass errors, shown as open diamonds on the "No error" row. Point size encodes the 16th–84th percentile width for sampled objects. The accounting panel retains all 101 catalogue-only objects.

Figure 4: Descriptive compatibility across seed families, spin/efficiency cases, merger boosts, and average accretion rates. Percentages are within admitted classes and are not completeness-corrected demographic estimates.

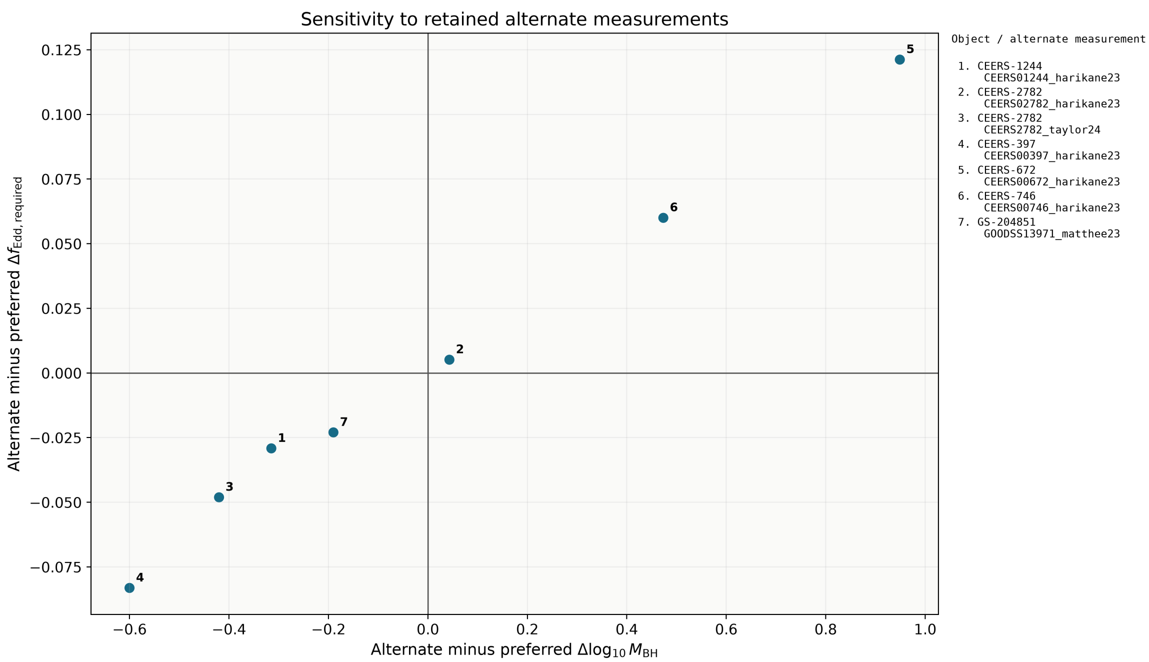

Figure 5: Sensitivity to seven retained alternate measurements. Numbered points are keyed to stable object and measurement identifiers.

5 Discussion

5.1 What the high-pressure tail establishes

The scientific question is which objects constrain growth histories and which constraints survive changes in the assumptions. All 14 reference point-estimate requirements above unity lie at $z = 6.6214$–10.603. The main growth tension is therefore concentrated in the early subset, not shared by every catalogue member. The primary 227-object comparison has 12 such requirements; the full 237-object set provides exploratory context across less comparable evidence.

The reference calculation identifies a small set of objects that are difficult to grow from $10^2\,M_\odot$ seeds at $f_{\rm Edd} \leq 1$ under the reference $\epsilon = 0.1$, $z_{\rm seed} = 30$ history. The result does not select a unique remedy. Earlier or heavier seeds, lower effective radiative efficiency, merger growth, or episodes above the reference accretion rate can each reduce the tension. The atlas is designed to expose these trade-offs for each object rather than collapse them into a single formation label.

This dependence can be quantified without changing the catalogue. At fixed seed mass, growth interval and merger factor, required $f_{\rm Edd}$ scales as $\epsilon/(1-\epsilon)$. The implemented ideal non-spinning thin-disk value $\epsilon = 1 - \sqrt{8/9} = 0.05719$ reduces every point requirement by a factor 0.54594 relative to $\epsilon = 0.1$: the largest becomes 0.793 and none exceeds unity. This sensitivity calculation does not infer actual spins or growth histories; it shows why the reference tail cannot establish a model-independent need for super-Eddington accretion.

The efficiency scaling itself is an algebraic consequence of the adopted model, not a new physical mechanism. The contribution is its application to a source-traceable catalogue, the assessment of measurement and model sensitivity, and the identification of objects for which independent observations can most change the constraints. A point requirement below unity does not exclude super-Eddington episodes, measure spin, or guarantee a sustainable fuel supply: near-continuous accretion can remain demanding even when the average is below Eddington. Scenario compatibility must not be interpreted as a unique history.

The most useful targets are therefore those whose placement is both demanding and stable to published mass uncertainty. Independent mass measurements, better separation of broad-line-region kinematics from scattering or outflows, and constraints on obscuration and lensing can change the physical interpretation more than another evaluation of the same analytic formula. The follow-up matrix makes that distinction explicit, but its heuristic ordering is not a calibrated calculation of expected information gain.

5.2 Value of the heterogeneous evidence layer

The 101 objects without eligible masses remain within project scope because the catalogue is a cross-paper evidence atlas as well as a growth calculator. Narrow-line, high-ionization, X-ray, and SED-selected candidates trace discovery channels that a mass-only broad-line catalogue would omit. They can be compared in redshift, evidence, survey, and follow-up need, but they are excluded from numerical growth claims until a suitable mass exists. This separation lets the atlas retain potentially important early accretors without converting qualitative evidence or model-dependent context into artificial precision.

5.3 Limitations

The dominant limitations are scientific rather than computational. The catalogue is a union of heterogeneous selections and cannot support pooled demographics without source-specific completeness models. Single-epoch virial masses share calibration and geometry uncertainties that are incompletely reported across papers. Two source-supported duplicate pairs have been reconciled without removing measurements. Three identity groups remain open: Baccus GDS_1210_9515 versus JADES GS-8083, GS-10013704 versus Scholtz 99671, and Scholtz 16745 versus 208643. The first requires target/spectral and preferred-mass reconciliation; the latter two require imaging and aperture checks. Unique-object and host counts remain provisional. These specific tasks cannot be replaced by wider matching thresholds. Finally, Eq. 4 compresses time-dependent accretion, feedback, mergers, and radiative-state changes into effective parameters. Its outputs should be read as conditional comparisons, not reconstructed histories. Reported-error probabilities condition on fixed growth parameters; marginalization requires justified parameter distributions and correlations, not just scenario maps. They are not seed-origin probabilities. The age relation neglects radiation; the supplementary $z_{\mathrm{seed}} = 3400$ plot is an extrapolation, not evidence for primordial formation or a physical radiation-era accretion history.

All admitted numerical mass/error fields have independent source checks; all 350 central redshifts and 323 available coordinate pairs are also checked against independent source evidence. Twenty-seven measurement rows lack coordinates. Redshift uncertainties, full source posteriors, and auxiliary observables are not exhaustively validated. The immediate priority is resolving the three identity groups and reassessing affected summaries. The literature cutoff is 3 September 2026; catalogue completeness refers to the declared source membership, not every JWST discovery. Demographic inference requires source-specific inclusion models and overlap corrections. The A2744-QSO1 direct-mass measurement [38] awaits a new dataset version under the frozen-membership policy.

6 Reproducibility and data products

The repository pins the direct Python 3.12 dependencies and provides five ordered notebooks that materialize catalogues, generate science tables, render figures, build the complete atlas, and verify the release. Canonical manifests record SHA-256 hashes for 85 v1, 461 v2, and 720 v3 artifacts. Verification checks strict nested version membership, catalogue cardinalities, numerical CSV reproduction, 10,000-draw uncertainty products, compatibility coverage, figure resolution, and both parameter-map panels for every object. Regenerated CSV values are compared with an independent committed baseline using the specified numerical tolerances. PNG dimensions and transparency must agree exactly; each RGB channel permits at most three levels of difference on the 0–255 scale, based on the measured cross-platform rasterization differences. No image resizing, alignment or averaging is used to accept a mismatch. The current result inventory contains 1,236 artifacts.

Independent source-cell verification performs 2,041 comparisons across three overlapping fixtures: 865 checks across 53 admitted rows from JADES, NEXUS, COSMOS-3D and Seven Wonders; 444 family-anchor checks; and 732 checks covering all 244 numerical masses and both error fields. This is a comparison count, not a count of unique cells. Other source fields retain representative coverage. These checks recovered omitted NEXUS line-luminosity errors, now preserved in source observables without changing masses or ranks. Numerical quadrature of the Friedmann age integral and Runge–Kutta growth integration using physical constants provide checks independent of the production closed-form functions. This supplements reproducibility; it is not an exhaustive independent re-audit of every measurement and interpretation across all 32 families.

Every v3 object has an f_{Edd}–mass panel and a seed-redshift–mass panel. The 237 eligible objects receive numerical products; the 101 others receive explicit status panels. Combined all-object compatibility and Monte Carlo atlases retain every label for close inspection. A supplementary full-assumption growth-track figure preserves the historical v1 grid of three seed masses, three f_{Edd} values, four constant efficiencies, and two merger boosts, for 72 curves.

7 Conclusions

We have constructed a reproducible atlas with 338 provisional object records of JWST-identified accreting black-hole objects and candidates at $z \geq 4$. The main results are:

- 237 objects have a numerically eligible mass and 101 remain explicit no-inference records;
- under the light-seed reference history, 14 objects require point-estimate $f_{\mathrm{Edd}} > 1$, ten remain above unity at the 16th percentile, and eight have $P(f_{\mathrm{Edd}} > 1) \geq 0.95$;
- UNCOVER-20466 places the strongest pressure on that history, while the new COSMOS3D-13852 entry ranks third by the composite navigation score and GS-20057765 is third by required f_{Edd};
- model compatibility is conditional on seed, efficiency, merger, and accretion assumptions, and the heterogeneous catalogue cannot support pooled demographic inference; lowering the fixed efficiency to the ideal non-spinning value removes all point-estimate requirements above unity; and
- complete provenance, identity history, uncertainty semantics, source caveats, follow-up priorities, and per-object visual products remain linked to the same frozen v3 dataset.

The immediate next step is to resolve the three remaining identity groups and reassess affected summaries. Within the present object-level scope, independent mass constraints and physically justified growth scenarios can then identify which conclusions are stable. Population inference would additionally require survey completeness, overlap corrections, and shared mass-calibration modelling.

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